FT_TRANSCENDENCE

Official Documentation — 42 School

1. Project Overview

ft_transcendence is the final project of the 42 School common core. It consists of building a fullstack web application to play a Rock-Paper-Scissors-Well game against an AI bot, with 3D graphics and an intelligent chatbot.

Member	Role
Lenny	DevOps and Cybersecurity
Elias	User Management
Maxence	Artificial Intelligence and RAG Chatbot
Lena	Frontend and the Rock Paper Scissors game
Capu	Graphics rendering with Blender and Three.js

2. Main Game — Rock, Paper, Scissors, Well

Moves

- Rock beats Scissors; loses to Paper and to the Well.
- Paper beats Rock and the Well; loses to Scissors.
- Scissors beats Paper; loses to Rock and to the Well.
- The Well beats Rock and Scissors; loses only to Paper.
- The Well is available from level 2 onward.
- If both players pick the same move, the round is a tie.

Scoring

- Winning a round gives +1 point. A loss or a tie gives nothing.
- From level 2, losing a round while playing the Well costs 2 points (score never drops below 0).
- A combo of three special winning moves in a row gives a +1 bonus point.

Winning a match

- Level 0: first to 5 points wins.
- Levels 1, 2 and 3: first to 10 points wins.

Graphics rendering

- The game uses Three.js for 3D rendering.
- The 3D assets were created with Blender by Capu.
- The game is compatible with Chrome and Firefox.

3. Levels and the AI Bot

- Level 0: classic Rock-Paper-Scissors, no Well, no combos, first to 5 points.
- Level 1: combos introduced (three Rocks, Papers or Scissors), no Well, first to 10.
- Level 2: the Well appears with its 2-point penalty, combos still apply, first to 10.
- Level 3: hardest level, Well plus extra combos and a giga-combo for three Wells, first to 10.

The AI bot has three difficulties: Easy (mostly random), Medium (default, predicts about 60% of the time) and Hard (mostly predicts). It never sees your current move; it predicts your next move from your move history and the result of the previous round, using a Markov transition combined with a win-stay / lose-shift heuristic.

4. Module — User Management (Elias)

Account creation

- Sign up via email + password.
- OAuth 42 authentication available.
- Unique username on the platform.
- A default avatar is assigned on creation.

Security

- JWT for user sessions.
- Passwords hashed with bcrypt.
- Two-factor authentication (2FA) available via TOTP.

5. Module — Artificial Intelligence (Maxence)

RAG Chatbot

- An intelligent chatbot answers questions about the game.
- Based on a RAG (Retrieval-Augmented Generation) system over the project documentation.
- Uses Ollama (nomic-embed-text) for embeddings and Groq (llama-3.3-70b-versatile) as the LLM.
- Answers in the same language as the question.
- Accessible only to logged-in users (protected by a JWT guard).

6. Module — Frontend and Game (Lena)

- Single Page Application built with Next.js / React.
- Modern, responsive design.
- Compatible with Chrome and Firefox.
- 3D rendering with Three.js and Blender assets (Capu).

7. Module — DevOps and Cybersecurity (Lenny)

- The whole project runs in containers.
- A single docker-compose file launches the entire application.
- Nginx as a reverse proxy with SSL/TLS (self-signed certificates for development).
- PostgreSQL (pgvector image) as the main database with Prisma as the ORM.
- Automatic migrations on startup.

8. Technical Stack

Technology	Usage
NestJS (TypeScript)	Backend
Next.js / React	Frontend
PostgreSQL + Prisma	Database
Three.js + Blender	3D rendering
Ollama + Groq + LangChain	AI / LLM / RAG
Docker + Nginx	Infrastructure
JWT + OAuth 42 + 2FA	Authentication

For any question, refer to the official subject on the 42 intranet.